ALGOTRADE

FIX 4.4 Trading API Specification

Table of Contents

- 0) Purpose & Background
- 1) Roles, Connectivity & Account Model
  - 1.1 Roles
  - 1.2 Account Model — Single Connection, Multiple Sub-accounts
  - 1.3 Session Schedule & Sequence Numbers
  - 1.4 Reconnection & Failover
- 2) Session Layer (Admin)
  - 2.1 Standard Header
  - 2.2 Logon (35=A)
  - 2.3 Logout (35=5)
  - 2.4 Mandatory FIX session support
- 3) Trading Messages (Application)
  - 3.1 New Order Single (35=D)
    - 3.1.1 Required fields
    - 3.1.2 Custom OrdType note
    - 3.1.3 TimeInForce (59) values
    - 3.1.4 Additional fields for derivatives
    - 3.1.5 Example — Equity
    - 3.1.6 Example — Derivatives
  - 3.2 Order Cancel Request (35=F)
- 4) Execution Report (35=8) — Fields, States & Mapping
  - 4.1 Fields expected on all Execution Reports
  - 4.2 Trade-specific fields
  - 4.3 Reject fields
    - 4.3.1 OrdRejReason (103) values
  - 4.4 ExecType (150) ↔ OrdStatus (39) Mapping
- 5) Order Cancel Reject (35=9)
  - 5.1 Required fields
  - 5.2 CxlRejReason (102) values
  - 5.3 Example
- 6) Business Message Reject (35=j)
  - 6.1 Required fields
  - 6.2 BusinessRejectReason (380) values
  - 6.3 Example
- 7) Order Lifecycles
- 8) Open Items & Questions for <Broker>
- 9) Public References
- Appendix A: Message Type Summary
- Appendix B: Tag Quick Reference

0) Purpose & Background

This document provides the **recommended FIX 4.4 specification** for <Broker>'s FIX trading server, covering message types, required fields, expected behaviors, and order lifecycle flows.

Algotrade provides this specification to support <Broker>'s FIX 4.4 implementation. FIX protocol connectivity enables **programmatic order submission with lower latency** compared to GUI-based workflows. This specification reflects common practices for FIX integrations in the Vietnamese market.

Algotrade will serve as a **FIX client to connect and jointly test** the implementation with <Broker>.

The specification covers **equities and derivatives** uniformly. Both product types use the same message types and session model; the only difference is a small number of additional tags for contract identification on derivatives orders (see section 3.1.4).

Covered in this document:

- Session: **Logon (A)**, **Logout (5)**, and standard FIX session-level messages
- Trading: **New Order Single (D)**, **Order Cancel Request (F)**
- Reporting: **Execution Report (8)** — order acknowledgements, cancels, fills, rejects, and expiry
- Reporting: **Order Cancel Reject (9)** — when a cancel request cannot be honored
- Reporting: **Business Message Reject (j)** — for unsupported or invalid application messages

Not covered (can be added later if needed):

- Order Cancel/Replace Request (G), mass cancel, allocation, position management, market data
- Full FIX 4.4 specification details — please refer to the official FIX documentation for all standard behaviors not explicitly covered here

1) Roles, Connectivity & Account Model

1.1 Roles

Aspect	Value
<Broker>	FIX **Acceptor (Server)**
Algotrade	FIX **Initiator (Client)**

1.2 Account Model — Single Connection, Multiple Sub-accounts

We recommend that the FIX server support **a single connection carrying orders for multiple sub-accounts**, across both equities and derivatives. This is a standard and scalable approach — it simplifies connectivity for both sides and avoids the overhead of managing separate sessions per account.

The **Account** tag (`1`) on every order identifies the specific sub-account. The FIX server should:

1. **Route orders** to the correct sub-account based on tag `1` .
2. **Echo back** the Account tag (`1`) on every Execution Report (`35=8`) and Order Cancel Reject (`35=9`) so that the client can attribute reports to the correct account.
3. Support **concurrent orders** from different sub-accounts on the same session without interference.

> **Note:** This is a one-to-many model (one FIX session → many accounts). It is the most common architecture for institutional FIX connectivity and is recommended over a one-to-one (one session per account) model.

1.3 Session Schedule & Sequence Numbers

Parameter	Recommended Value
Session start	Suggest: before market open (e.g., 08:30 ICT / 01:30 UTC)
Session end	Suggest: after market close (e.g., 15:00 ICT / 08:00 UTC)
Sequence number reset	Daily at session start, via `141=Y` (ResetSeqNumFlag) on Logon
Heartbeat interval	30 seconds (configurable on Logon via tag `108`)

1.4 Reconnection & Failover

- If the TCP connection is lost, the client will reconnect and send a new Logon.
- If `141=Y` is **not** set, both sides should honor existing sequence numbers and use **ResendRequest (35=2)** / **SequenceReset (35=4)** to recover any gaps.
- If `141=Y` is set, sequence numbers reset to 1 on both sides.
- **Working orders** should remain active on the exchange during a disconnect. The server should buffer any Execution Reports generated while the client is disconnected and deliver them upon reconnection (via the standard gap-fill mechanism).

- **Reconnection rate:** the client will not attempt more than 1 reconnection per 5 seconds to avoid overwhelming the server.

2) Session Layer (Admin)

2.1 Standard Header (required on all messages)

Tag	Field	Req	Notes
8	BeginString	Y	`FIX.4.4`
9	BodyLength	Y	Standard FIX
35	MsgType	Y	Message type
34	MsgSeqNum	Y	Monotonic per session
49	SenderCompID	Y	Sender's CompID (Algotrade when sending)
56	TargetCompID	Y	Receiver's CompID (<Broker> when sending)
52	SendingTime	Y	UTC timestamp
10	CheckSum	Y	Standard FIX

2.2 Logon (35=A)

Algotrade (Initiator) sends Logon to start a session. **<Broker> (Acceptor)** responds with Logon.

Required fields

- `35=A`
- `98=0` (EncryptMethod = None)
- `108=<secs>` (HeartBtInt)

Optional / by agreement

- `141=Y` (ResetSeqNumFlag) — only when both sides agree to reset sequence numbers
- Credentials — if <Broker> requires password auth:
  - `553` (Username)
  - `554` (Password)

Example — Logon with credentials and seq reset (illustrative)

```
8=FIX.4.4|35=A|49=ALGOTRADE|56=BROKER|34=1|52=20260101-00:00:00.000|
98=0|108=30|141=Y|553=algotrade_user|554=p@ssw0rd|10=000|
```

Example — Logon response from <Broker>

```
8=FIX.4.4|35=A|49=BROKER|56=ALGOTRADE|34=1|52=20260101-00:00:00.050|
98=0|108=30|141=Y|10=000|
```

2.3 Logout (35=5)

Either party may initiate logout.

- Send `35=5`, optional `58` (Text) for reason
- The peer should respond with a Logout if it has not already sent one

Example — Logout initiated by Algotrade

```
8=FIX.4.4|35=5|49=ALGOTRADE|56=BROKER|34=100|52=20260101-08:00:00.000|
58=End of trading day|10=000|
```

Example — Logout response from <Broker>

```
8=FIX.4.4|35=5|49=BROKER|56=ALGOTRADE|34=95|52=20260101-08:00:00.050|
58=Goodbye|10=000|
```

2.4 Mandatory FIX session support

Both sides must support standard FIX 4.4 session messages:

- Heartbeat (35=0), TestRequest (35=1)
- ResendRequest (35=2), SequenceReset (35=4)
- Reject (35=3) — for protocol-level errors (malformed messages, invalid tags, etc.)

> **Note on examples:** Throughout this document, `|` is used in place of SOH (`\x01`) for readability. `BodyLength (9)` is omitted and `CheckSum (10)` is shown as `000`. In production, all must conform to the FIX specification.

3) Trading Messages (Application)

> Unless otherwise stated, all timestamps are **UTC**.

3.1 New Order Single (35=D)

Used to place a new order for equities or derivatives.

3.1.1 Required fields

Tag	Field	Req	Notes
35	MsgType	Y	`D`
11	ClOrdID	Y	Client order id. **Must be unique** per trading day (recommend ≤ 30 chars).
1	Account	Y	Sub-account identifier (see section 1.2). Multiple accounts may submit orders on the same FIX session.
207	SecurityExchange	Y	`HSX` / `HNX` / `UPCOM`
55	Symbol	Y	Ticker only (e.g., `MWG`) unless later agreed otherwise
54	Side	Y	`1=Buy`, `2=Sell`
38	OrderQty	Y	Quantity
40	OrdType	Y	`1=Market`, `2=Limit`, `K=MTL` *(custom — see note below)*
44	Price	C	**Required** when `40=2` (Limit); omitted otherwise
59	TimeInForce	Y	See below
60	TransactTime	Y	Order creation time

3.1.2 Custom OrdType note

`OrdType K = MTL (Market-to-Limit)` is a custom extension specific to Vietnamese exchanges. It is not part of the standard FIX 4.4 OrdType enumeration. Both sides agree to support this value.

3.1.3 TimeInForce (59) values

The following values cover all standard Vietnamese order types (LO, ATO, ATC, MAK, MOK). All values are standard FIX 4.4. Availability may vary between equities and derivatives sessions on the exchange.

Value	Name	VN Order Type	Notes
`0`	Day	LO, MTL	Default for Limit / MTL
`2`	OPG (At the Opening)	ATO	

Value	Name	VN Order Type	Notes
`3`	IOC (Immediate or Cancel)	MAK	Default for Market
`4`	FOK (Fill or Kill)	MOK	
`7`	AtTheClose	ATC	

3.1.4 Additional fields for derivatives

For derivatives orders (futures, options), the following fields should be included in addition to those in 3.1.1:

Tag	Field	Req	Notes
167	SecurityType	Y	`FUT` for futures; `OPT` for options (if applicable)
200	MaturityMonthYear	C	Format `YYYYMM` — required if `55` alone is ambiguous
48	SecurityID	C	Alternative to `55` if <Broker> uses a numeric ID scheme
22	SecurityIDSource	C	Required when `48` is used (e.g., `8` = Exchange Symbol)

> **Question for <Broker>:** Please confirm the preferred contract identification scheme for derivatives — Symbol-based (`55` , e.g., `VN30F2603`) or SecurityID-based (`48` + `22`).

3.1.5 Example — Equity (Limit Order, Day)

```
8=FIX.4.4|35=D|49=ALGOTRADE|56=BROKER|34=21|52=20260101-01:00:00.000|
1=ACC001|11=CL000001|207=HSX|55=MWG|54=1|38=100|40=2|44=50000|59=0|60=20260
101-01:00:00.000|10=000|
```

3.1.6 Example — Derivatives (Futures, Limit Order)

```
8=FIX.4.4|35=D|49=ALGOTRADE|56=BROKER|34=100|52=20260315-02:00:00.000|
1=ACC002|11=CL000050|167=FUT|207=HNX|55=VN30F2606|54=1|38=10|40=2|44=1350|5
9=0|60=20260315-02:00:00.000|10=000|
```

3.2 Order Cancel Request (35=F)

Requests cancellation of the remaining quantity of a working order.

Required fields

Tag	Field	Req	Notes

Tag	Field	Req	Notes
35	MsgType	Y	`F`
11	ClOrdID	Y	**New** cancel request id (unique)
1	Account	Y	Same as original order
37	OrderID	Y	Order id assigned by <Broker> (from Execution Report)
41	OrigClOrdID	Y	Original ClOrdID of the order being cancelled
207	SecurityExchange	Y	Exchange code
55	Symbol	Y	Ticker/contract symbol
54	Side	Y	Same as original order
60	TransactTime	Y	Cancel request time

Example

```
8=FIX.4.4|35=F|49=ALGOTRADE|56=BROKER|34=40|52=20260101-01:05:00.000|
11=CXL000001|1=ACC001|37=123456|41=CL000001|207=HSX|55=MWG|54=1|60=20260101
-01:05:00.000|10=000|
```

4) Execution Report (35=8) — Fields, States & Mapping

The FIX server sends Execution Reports to confirm order state changes, trades, cancels, and rejects.

4.1 Fields expected on all Execution Reports

Tag	Field	Req	Notes
35	MsgType	Y	`8`
150	ExecType	Y	Execution type (see mapping in 4.4)
39	OrdStatus	Y	Order status (see mapping in 4.4)
17	ExecID	Y	Unique execution id (unique across the session)
37	OrderID	Y	<Broker>-assigned order id
11	ClOrdID	Y	Client ClOrdID (for cancel flow: the cancel request's ClOrdID)
41	OrigClOrdID	C	Required for cancel-related reports
1	Account	Y	Sub-account identifier (echoed from the original order)
55	Symbol	Y	Ticker / contract symbol
207	SecurityExchange	Y	Exchange code
54	Side	Y	`1` / `2`
38	OrderQty	Y	Original order quantity
40	OrdType	Y	As received
59	TimeInForce	Y	As received
44	Price	C	Required when `OrdType=2` (Limit)
151	LeavesQty	Y	Remaining open quantity
14	CumQty	Y	Cumulative filled quantity
6	AvgPx	Y	Average fill price (`0` if no fills yet)
60	TransactTime	Y	Execution or event timestamp

4.2 Trade-specific fields (required when a fill occurs)

When `150=F` (Trade) is sent:

- `31` (LastPx) **required**
- `32` (LastQty) **required**

4.3 Reject fields (required when `150=8`)

When an order is rejected:

- `150=8` (Rejected), `39=8` (Rejected)

- `103` (OrdRejReason) **required** — see values below
- `58` (Text) **required** — human-readable reason

4.3.1 OrdRejReason (103) values

Value	Meaning
`0`	Broker / exchange option
`1`	Unknown symbol
`2`	Exchange closed
`3`	Order exceeds limit
`5`	Unknown order
`6`	Duplicate order (ClOrdID)
`13`	Incorrect quantity
`99`	Other

> <Broker> may define additional values beyond the standard set above. We would appreciate if any custom rejection reasons are documented and shared so that Algotrade can handle them correctly.

4.4 ExecType (150) ↔ OrdStatus (39) Mapping

The following table defines the expected combinations. We recommend adhering to this standard mapping for consistency.

Scenario	ExecType (150)	OrdStatus (39)	LeavesQty	CumQty
Pending New (broker acknowledged)	`A`	`A`	= OrderQty	0
New (exchange confirmed)	`0`	`0`	= OrderQty	0
Partial Fill	`F`	`1`	> 0	> 0
Full Fill	`F`	`2`	0	= OrderQty
Pending Cancel	`6`	`6`	unchanged	unchanged
Cancelled	`4`	`4`	0	unchanged
Rejected	`8`	`8`	0	0
Expired (IOC/FOK/end-of-day)	`C`	`C`	0	unchanged

Important notes:

- **OrdStatus precedence:** If an order is partially filled *and* a cancel is pending, the OrdStatus must be `6` (Pending Cancel), not `1` (Partially Filled). Pending states always take precedence per the FIX specification.

- **Pending Cancel is optional to send:** If the cancellation is processed immediately, the server may skip the Pending Cancel step and send `150=4, 39=4` directly.
- **Expired state:** IOC or FOK orders that do not fully fill should receive `150=C, 39=C` as the terminal report. Day orders that are still open at end-of-day should also receive an Expired or Cancelled report.

5) Order Cancel Reject (35=9)

When <Broker> cannot honor a cancel request (e.g., order already filled, unknown order, too late to cancel), the server should respond with an **Order Cancel Reject** instead of an Execution Report.

5.1 Required fields

Tag	Field	Req	Notes
35	MsgType	Y	`9`
11	ClOrdID	Y	The ClOrdID from the cancel request
37	OrderID	Y	<Broker>-assigned order id
41	OrigClOrdID	Y	Original order's ClOrdID
1	Account	Y	Sub-account identifier
39	OrdStatus	Y	Current status of the order (e.g., `0`, `1`, `2`)
434	CxlRejResponseTo	Y	`1` = Order Cancel Request
102	CxlRejReason	Y	See values below
58	Text	Y	Human-readable explanation
60	TransactTime	Y	Timestamp

5.2 CxlRejReason (102) values

Value	Meaning
`0`	Too late to cancel
`1`	Unknown order
`2`	Broker / exchange option
`3`	Order already in Pending Cancel or Pending Replace status
`6`	Duplicate ClOrdID on cancel request
`99`	Other

5.3 Example

```
8=FIX.4.4|35=9|49=BROKER|56=ALGOTRADE|34=45|52=20260101-01:06:00.000|
11=CXL000001|37=123456|41=CL000001|1=ACC001|39=2|434=1|102=0|58=Too late to
cancel, order already filled|60=20260101-01:06:00.000|10=000|
```

6) Business Message Reject (35=j)

The FIX server should send a Business Message Reject when it receives a structurally valid FIX message that cannot be processed for business-level reasons (e.g., unsupported `MsgType`, application not available). This is distinct from Session-level Reject (`35=3`), which handles protocol-level errors.

6.1 Required fields

Tag	Field	Req	Notes
35	MsgType	Y	`j`
45	RefSeqNum	Y	MsgSeqNum of the rejected message
372	RefMsgType	Y	MsgType of the rejected message
380	BusinessRejectReason	Y	See values below
58	Text	Y	Human-readable reason

6.2 BusinessRejectReason (380) values

Value	Meaning
`0`	Other
`1`	Unknown ID
`2`	Unknown Security
`3`	Unsupported Message Type
`4`	Application not available
`5`	Conditionally required field missing
`6`	Not authorized

6.3 Example

```
8=FIX.4.4|35=j|49=BROKER|56=ALGOTRADE|34=50|52=20260101-01:10:00.000|
45=48|372=D|380=6|58=Account not authorized to trade on HNX|10=000|
```

7) Order Lifecycles

Full message examples for each scenario. All examples use: Buy MWG on HSX, Account `ACC001`. `BodyLength (9)` omitted, `CheckSum (10)` shown as `000`.

7.1 New Order → Accepted

Step 1. Algotrade → <Broker>: New Order Single

```
8=FIX.4.4|35=D|49=ALGOTRADE|56=BROKER|34=2|52=20260101-01:00:00.000|
1=ACC001|11=CL000001|207=HSX|55=MWG|54=1|38=200|40=2|44=50000|59=0|60=20260
101-01:00:00.000|10=000|
```

Step 2. <Broker> → Algotrade: Pending New (`150=A, 39=A`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=2|52=20260101-01:00:00.100|
150=A|39=A|17=EXEC001|37=ORD0001|11=CL000001|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=2|44=50000|59=0|
151=200|14=0|6=0|60=20260101-01:00:00.100|10=000|
```

Step 3. <Broker> → Algotrade: New (`150=0, 39=0`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=3|52=20260101-01:00:00.200|
150=0|39=0|17=EXEC002|37=ORD0001|11=CL000001|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=2|44=50000|59=0|
151=200|14=0|6=0|60=20260101-01:00:00.200|10=000|
```

7.2 New Order → Rejected

Step 1. Algotrade → <Broker>: New Order Single

```
8=FIX.4.4|35=D|49=ALGOTRADE|56=BROKER|34=2|52=20260101-01:00:00.000|
1=ACC001|11=CL000002|207=HSX|55=XYZ|54=1|38=100|40=2|44=10000|59=0|60=20260
101-01:00:00.000|10=000|
```

Step 2. <Broker> → Algotrade: Rejected (`150=8, 39=8`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=2|52=20260101-01:00:00.100|
150=8|39=8|17=EXEC010|37=ORD0002|11=CL000002|1=ACC001|55=XYZ|207=HSX|54=1|3
8=100|40=2|44=10000|59=0|
151=0|14=0|6=0|103=1|58=Unknown symbol|60=20260101-01:00:00.100|10=000|
```

7.3 New Order → Partial Fill → Full Fill

Step 1. Algotrade → <Broker>: New Order Single (Buy 200 MWG @ 50,000)

```
8=FIX.4.4|35=D|49=ALGOTRADE|56=BROKER|34=2|52=20260101-01:00:00.000|
1=ACC001|11=CL000001|207=HSX|55=MWG|54=1|38=200|40=2|44=50000|59=0|60=20260
101-01:00:00.000|10=000|
```

Step 2. <Broker> → Algotrade: Pending New (`150=A, 39=A`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=2|52=20260101-01:00:00.100|
150=A|39=A|17=EXEC001|37=ORD0001|11=CL000001|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=2|44=50000|59=0|
151=200|14=0|6=0|60=20260101-01:00:00.100|10=000|
```

Step 3. <Broker> → Algotrade: New (`150=0, 39=0`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=3|52=20260101-01:00:00.200|
150=0|39=0|17=EXEC002|37=ORD0001|11=CL000001|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=2|44=50000|59=0|
151=200|14=0|6=0|60=20260101-01:00:00.200|10=000|
```

Step 4. <Broker> → Algotrade: Partial Fill (`150=F, 39=1`) — filled 100 @ 50,000

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=4|52=20260101-01:01:00.000|
150=F|39=1|17=EXEC003|37=ORD0001|11=CL000001|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=2|44=50000|59=0|
32=100|31=50000|151=100|14=100|6=50000|60=20260101-01:01:00.000|10=000|
```

Step 5. <Broker> → Algotrade: Full Fill (`150=F, 39=2`) — filled 100 @ 50,500

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=5|52=20260101-01:02:00.000|
150=F|39=2|17=EXEC004|37=ORD0001|11=CL000001|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=2|44=50000|59=0|
32=100|31=50500|151=0|14=200|6=50250|60=20260101-01:02:00.000|10=000|
```

7.4 Cancel — Success

Assumes order `CL000001` is in New state (`39=0`) with `OrderID=ORD0001` .

Step 1. Algotrade → <Broker>: Order Cancel Request

```
8=FIX.4.4|35=F|49=ALGOTRADE|56=BROKER|34=5|52=20260101-01:05:00.000|
11=CXL000001|1=ACC001|37=ORD0001|41=CL000001|207=HSX|55=MWG|54=1|60=2026010
1-01:05:00.000|10=000|
```

Step 2. <Broker> → Algotrade: Pending Cancel (`150=6, 39=6`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=6|52=20260101-01:05:00.100|
150=6|39=6|17=EXEC005|37=ORD0001|11=CXL000001|41=CL000001|1=ACC001|55=MWG|2
07=HSX|54=1|38=200|40=2|44=50000|59=0|
151=200|14=0|6=0|60=20260101-01:05:00.100|10=000|
```

Step 3. <Broker> → Algotrade: Cancelled (`150=4, 39=4`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=7|52=20260101-01:05:00.200|
150=4|39=4|17=EXEC006|37=ORD0001|11=CXL000001|41=CL000001|1=ACC001|55=MWG|2
07=HSX|54=1|38=200|40=2|44=50000|59=0|
151=0|14=0|6=0|60=20260101-01:05:00.200|10=000|
```

> Step 2 (Pending Cancel) may be skipped if the cancel is processed immediately.

7.5 Cancel — Rejected (order already filled)

Step 1. Algotrade → <Broker>: Order Cancel Request

```
8=FIX.4.4|35=F|49=ALGOTRADE|56=BROKER|34=8|52=20260101-01:06:00.000|
11=CXL000002|1=ACC001|37=ORD0001|41=CL000001|207=HSX|55=MWG|54=1|60=2026010
1-01:06:00.000|10=000|
```

Step 2. <Broker> → Algotrade: Order Cancel Reject (`35=9`)

```
8=FIX.4.4|35=9|49=BROKER|56=ALGOTRADE|34=9|52=20260101-01:06:00.100|
11=CXL000002|37=ORD0001|41=CL000001|1=ACC001|39=2|434=1|102=0|58=Too late
to cancel, order already filled|
60=20260101-01:06:00.100|10=000|
```

7.6 Partial Fill → Cancel Remaining

Step 1. Algotrade → <Broker>: New Order Single (Buy 200 MWG @ 50,000)

```
8=FIX.4.4|35=D|49=ALGOTRADE|56=BROKER|34=2|52=20260101-01:00:00.000|
1=ACC001|11=CL000001|207=HSX|55=MWG|54=1|38=200|40=2|44=50000|59=0|60=20260
101-01:00:00.000|10=000|
```

Step 2. <Broker> → Algotrade: Pending New (`150=A, 39=A`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=2|52=20260101-01:00:00.100|
150=A|39=A|17=EXEC001|37=ORD0001|11=CL000001|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=2|44=50000|59=0|
151=200|14=0|6=0|60=20260101-01:00:00.100|10=000|
```

Step 3. <Broker> → Algotrade: New (`150=0, 39=0`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=3|52=20260101-01:00:00.200|
150=0|39=0|17=EXEC002|37=ORD0001|11=CL000001|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=2|44=50000|59=0|
151=200|14=0|6=0|60=20260101-01:00:00.200|10=000|
```

Step 4. <Broker> → Algotrade: Partial Fill (`150=F, 39=1`) — filled 100 @ 50,000

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=4|52=20260101-01:01:00.000|
150=F|39=1|17=EXEC003|37=ORD0001|11=CL000001|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=2|44=50000|59=0|
32=100|31=50000|151=100|14=100|6=50000|60=20260101-01:01:00.000|10=000|
```

Step 5. Algotrade → <Broker>: Order Cancel Request

```
8=FIX.4.4|35=F|49=ALGOTRADE|56=BROKER|34=5|52=20260101-01:02:00.000|
11=CXL000001|1=ACC001|37=ORD0001|41=CL000001|207=HSX|55=MWG|54=1|60=2026010
1-01:02:00.000|10=000|
```

Step 6. <Broker> → Algotrade: Cancelled (`150=4, 39=4`) — 100 filled, 100 cancelled

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=5|52=20260101-01:02:00.100|
150=4|39=4|17=EXEC007|37=ORD0001|11=CXL000001|41=CL000001|1=ACC001|55=MWG|2
07=HSX|54=1|38=200|40=2|44=50000|59=0|
151=0|14=100|6=50000|60=20260101-01:02:00.100|10=000|
```

7.7 IOC Order → Expired (no fill)

Step 1. Algotrade → <Broker>: New Order Single (IOC Market, 100 shares)

```
8=FIX.4.4|35=D|49=ALGOTRADE|56=BROKER|34=2|52=20260101-01:00:00.000|
1=ACC001|11=CL000003|207=HSX|55=MWG|54=1|38=100|40=1|59=3|60=20260101-
01:00:00.000|10=000|
```

Step 2. <Broker> → Algotrade: Pending New (`150=A, 39=A`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=2|52=20260101-01:00:00.100|
150=A|39=A|17=EXEC020|37=ORD0003|11=CL000003|1=ACC001|55=MWG|207=HSX|54=1|3
8=100|40=1|59=3|
151=100|14=0|6=0|60=20260101-01:00:00.100|10=000|
```

Step 3. <Broker> → Algotrade: Expired (`150=C, 39=C`)

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=3|52=20260101-01:00:00.200|
150=C|39=C|17=EXEC021|37=ORD0003|11=CL000003|1=ACC001|55=MWG|207=HSX|54=1|3
8=100|40=1|59=3|
151=0|14=0|6=0|60=20260101-01:00:00.200|10=000|
```

7.8 IOC Order → Partial Fill → Expired

Step 1. Algotrade → <Broker>: New Order Single (IOC Market, 200 shares)

```
8=FIX.4.4|35=D|49=ALGOTRADE|56=BROKER|34=2|52=20260101-01:00:00.000|
1=ACC001|11=CL000004|207=HSX|55=MWG|54=1|38=200|40=1|59=3|60=20260101-
01:00:00.000|10=000|
```

Step 2. <Broker> → Algotrade: Partial Fill (`150=F, 39=1`) — filled 80 @ 50,000

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=2|52=20260101-01:00:00.100|
150=F|39=1|17=EXEC030|37=ORD0004|11=CL000004|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=1|59=3|
32=80|31=50000|151=120|14=80|6=50000|60=20260101-01:00:00.100|10=000|
```

Step 3. <Broker> → Algotrade: Expired (`150=C, 39=C`) — remaining 120 expired

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=3|52=20260101-01:00:00.200|
150=C|39=C|17=EXEC031|37=ORD0004|11=CL000004|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=1|59=3|
151=0|14=80|6=50000|60=20260101-01:00:00.200|10=000|
```

7.9 Unsolicited Cancel (exchange-initiated)

Exchange cancels order without client request (e.g., trading halt, regulatory action).

```
8=FIX.4.4|35=8|49=BROKER|56=ALGOTRADE|34=10|52=20260101-02:00:00.000|
150=4|39=4|17=EXEC040|37=ORD0001|11=CL000001|1=ACC001|55=MWG|207=HSX|54=1|3
8=200|40=2|44=50000|59=0|
151=0|14=0|6=0|58=Trading halt - order cancelled by exchange|60=20260101-
02:00:00.000|10=000|
```

Algotrade identifies this as unsolicited because no preceding cancel request (`35=F`) was sent.

8) Open Items & Questions for <Broker>

We would be happy to discuss any of the following items. Confirming these early will help ensure a smooth implementation and testing process.

#	Item	Status
1	UAT and Production endpoints (host, port, CompIDs)	Pending
2	Session schedule (session start/end times)	Pending
3	SSL/TLS requirement on transport layer	Pending
4	Authentication method — Username/Password (`553` / `554`) on Logon?	Pending
5	Derivatives contract identification scheme (`55` vs `48` + `22`)	Pending
6	Custom OrdRejReason or CxlRejReason values the server plans to use	Pending
7	Will Execution Reports include fee information? (tags `136/137/139`)	Pending
8	Maximum message rate / throttling limits	Pending
9	Behavior of working orders during disconnect — remain active?	Pending
10	End-of-day handling — Expired (`39=C`) or Cancelled (`39=4`) for unfilled Day orders?	Pending

9) Public References

- FIX 4.4 order state changes: https://www.fixtrading.org/online-specification/order-state-changes/
- FIX 4.4 field dictionary: https://www.onixs.biz/fix-dictionary/4.4/fields_by_tag.html

Appendix A: Message Type Summary

MsgType	Name	Direction	Description
`A`	Logon	Both	Session initiation
`5`	Logout	Both	Session termination
`0`	Heartbeat	Both	Connection keep-alive
`1`	TestRequest	Both	Solicit heartbeat
`2`	ResendRequest	Both	Request retransmission of messages
`4`	SequenceReset	Both	Reset sequence numbers or gap-fill
`3`	Reject	Both	Session-level message rejection
`D`	NewOrderSingle	Algotrade → <Broker>	Submit a new order
`F`	OrderCancelRequest	Algotrade → <Broker>	Request order cancellation
`8`	ExecutionReport	<Broker> → Algotrade	Order status, fills, cancels, rejects
`9`	OrderCancelReject	<Broker> → Algotrade	Cancel request rejected
`j`	BusinessMessageReject	<Broker> → Algotrade	Application-level rejection

Appendix B: Tag Quick Reference

All application-level tags used in this specification:

Tag	Field	Used In
1	Account	D, F, 8, 9
6	AvgPx	8
11	ClOrdID	D, F, 8, 9
14	CumQty	8
17	ExecID	8
22	SecurityIDSource	D (derivatives)
31	LastPx	8 (fills)
32	LastQty	8 (fills)
37	OrderID	F, 8, 9
38	OrderQty	D, 8
39	OrdStatus	8, 9
40	OrdType	D, 8
41	OrigClOrdID	F, 8, 9

Tag	Field	Used In
44	Price	D, 8
45	RefSeqNum	j
48	SecurityID	D (derivatives)
54	Side	D, F, 8
55	Symbol	D, F, 8
58	Text	8, 9, j
59	TimeInForce	D, 8
60	TransactTime	D, F, 8, 9
102	CxlRejReason	9
103	OrdRejReason	8
150	ExecType	8
151	LeavesQty	8
167	SecurityType	D (derivatives)
200	MaturityMonthYear	D (derivatives)
207	SecurityExchange	D, F, 8
372	RefMsgType	j
380	BusinessRejectReason	j
434	CxlRejResponseTo	9